Central Limit Theorem and Convergence of Random Sequences

Lecture 9

Nov. 13, 18 and 20, 2025

Outline

Central Limit Theorem

- Let $X_1, X_2, \cdots, X_n$ be i.i.d. random variables with finite mean $E\{X\}$ and variance σ_X^2. Define the standardized sum as

$$Z_n = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \underbrace{\frac{X_k - E\{X\}}{\sigma_X}}_{=Y_k} .$$

  CLT states that as $n \to \infty$, the characteristic function (CF) $\Phi_{Z_n}(\omega) = E\{\exp(j\omega Z_n)\}$ of Z_n tends to the CF of $\mathcal{N}(0,1)$.

- Denote the CF of Y_k by $\Phi_Y(\omega) = E\{\exp(j\omega Y_k)\}$, $(j = \sqrt{-1})$. Then

$$\begin{aligned}\Phi_{Z_n}(\omega) =& E\{\exp(j\omega Z_n)\} = \prod_{k=1}^{n} E\{\exp(j\omega Y_k/\sqrt{n})\} = \prod_{k=1}^{n} \Phi_Y(\omega/\sqrt{n}) \\ =& \left(\Phi_Y(\frac{\omega}{\sqrt{n}})\right)^n\end{aligned}$$

- Taylor series of a function $g(u)$ around $u = u_0$ is

$$g(u) = g(u_0) + \sum_{\ell=1}^{L-1} g^{(\ell)}(u_0)\,\frac{(u-u_0)^\ell}{\ell!} + g^{(L)}(\bar{u})\,\frac{(u-u_0)^L}{L!}$$

for some $\bar{u} \in [u_0, u]$, where

$$g^{(\ell)}(u_0) = \left.\frac{d^\ell g(u)}{du^\ell}\right|_{u=u_0} = \ell\text{th derivative evaluated at } u_0$$

- Apply this to $\Phi_Y(\omega/\sqrt{n})$ around $\omega = 0$. Note that

$$\begin{aligned} \frac{d^\ell e^{j\omega y/\sqrt{n}}}{d\omega^\ell} =& (jy/\sqrt{n})^\ell e^{j\omega y/\sqrt{n}} \\ \Rightarrow \ \Phi_Y^{(\ell)}(0) =& E\{(jY/\sqrt{n})^\ell e^0\} = \left(\frac{j}{\sqrt{n}}\right)^\ell E\{Y^\ell\} \end{aligned}$$

- Therefore,

$$\Phi_Y(0) = E\{1\} = 1,\ \Phi_Y^{(1)}(0) = (j/\sqrt{n})E\{Y\} = 0\,,$$
$$\Phi_Y^{(2)}(0) = (j^2/n)E\{Y^2\} = -\frac{1}{n}, \quad \cdots$$

- For large n, we approximate $\Phi_Y(\omega/\sqrt{n})$ by first three terms in its Taylor series expansion around $\omega = 0$:

$$\Phi_Y(\omega/\sqrt{n}) \approx 1 - \frac{\omega^2}{2n}.$$

- Since $\Phi_{Z_n}(\omega) = (\Phi_Y(\frac{\omega}{\sqrt{n}}))^n$, we have

$$\ln \Phi_{Z_n}(\omega) = n \ln \Phi_Y(\omega/\sqrt{n}) \approx n \ln \left(1 - \frac{\omega^2}{2n}\right)$$

- A Taylor series expansion of $\ln(1+\delta)$ around $\delta = 0$ is

$$\ln(1+\delta) = \delta - \frac{\delta^2}{2} + \frac{\delta^3}{3} - \cdots \Rightarrow \ln(1+\delta) \approx \delta \ \text{ for } \ |\delta| \ll 1\,.$$

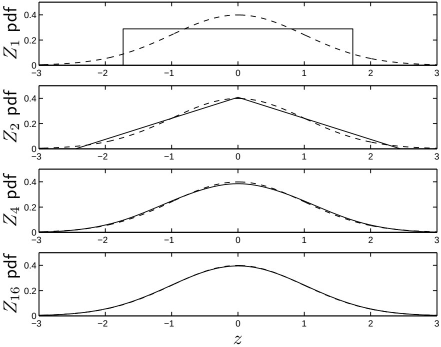
- Therefore, for "large" n,

$$\ln \Phi_{Z_n}(\omega) \approx n\left(-\frac{\omega^2}{2n}\right) = -\frac{\omega^2}{2} \ \Rightarrow \ \Phi_{Z_n}(\omega) \approx \exp\left(-\frac{\omega^2}{2}\right).$$

- Recall that if $Y \sim \mathcal{N}(0,1)$, its CF is $\Phi_Y(\omega) = \exp(-\frac{\omega^2}{2})$.
- It can be shown that convergence of CF implies convergence of the cumulative distribution function. But it does not necessarily imply convergence of the pdf.

Examples

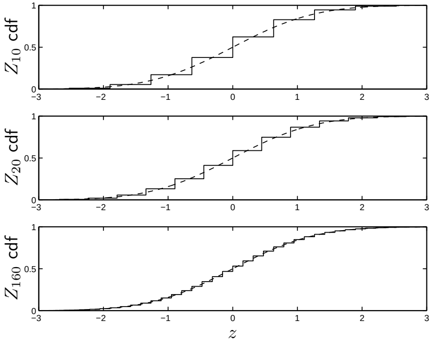
Example 1. Let $X_1, X_2, \cdots, X_n$ be i.i.d. random variables, uniformly distributed over $[-1.7, 1.7]$. The following plots show the pdf of Z_n for $n = 1, 2, 4, 16$. Note how fast the pdf of Z_n becomes close to Gaussian



Example 2. Let $X_1, X_2, \cdots, X_n$ be i.i.d. Bernoulli random variables with $p = 0.5$. Let $(E\{X_k\} = 0.5$ and $\sigma_X = 0.5)$

$$Z_n = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \frac{X_k - 0.5}{0.5}.$$

The following plots show the cdf of Z_n for $n = 10, 20, 160$. Of course Z_n is discrete, thus has no pdf. Its cdf converges to the Gaussian cdf.



Motivation

- One of the key questions in statistics is how to estimate the statistics of an r.v., e.g., its mean, variance, distribution, etc.
- To estimate such a statistic, we collect samples and use an estimator in the form of a sample average .
  - How good is the estimator? Does it "converge" to the true statistic?
  - How many samples do we need to ensure with some confidence that we are within a certain range of the true value of the statistic?
- These questions are answered using probability theory. The answers are called limit theorems.
- Example: Suppose we are given a coin with unknown bias p. To estimate the bias we flip the coin n times and compute the relative frequency of occurrence of heads n_H/n.
  - Does $n_H/n \to p$?
  - How many coin flips do we need?

The Sample Mean

- We can cast the above example as an example of estimating the mean of a r.v.. Suppose X is a r.v. with finite but unknown mean $E\{X\}$ (e.g., for the coin flipping example $X \sim \mathit{Bernoulli}(p)$ and we want to estimate $E\{X\} = p$).
- To estimate the mean, we collect $X_1, X_2, \cdots, X_n$ i.i.d. samples drawn according to the same distribution as X and compute the sample mean

$$S_n = \frac{1}{n} \sum_{i=1}^{n} X_i \,.$$

  - Does $S_n \to E\{X\}$ as $n \to \infty$?
  - What do we mean by convergence here?
  - How large should n be?

- Mean and variance of S_n: We have

$$E\{S_n\} = \frac{1}{n}\sum_{i=1}^{n} E\{X_i\} = E\{X\}$$

$$\text{var}(S_n) = E\{(S_n - E\{X\})^2\} = \frac{1}{n^2}\text{var}(\sum_{i=1}^{n} X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\text{var}(X_i) = \frac{\sigma_X^2}{n}$$

- Thus, $\text{var}(S_n) \to 0$ as $n \to \infty$.
- This result is an example of a limit theorem. It says that the sample mean converges in mean square to the true mean of the r.v.
- We will prove another limit theorem called the Weak Law of Large Numbers using this result.
- We then answer the question of how many samples are needed using the Central Limit Theorem.

The Weak Law of Large Numbers (WLLN)

- The WLLN states that if $X_1, X_2, \cdots, X_n$ is a sequence of i.i.d. r.v.'s with finite mean $E\{X\}$ and variance σ_X^2, then for any $\epsilon > 0$,

$$\lim_{n\to\infty} P(|S_n - E\{X\}| > \epsilon) = 0$$

- Proof: By the Chebyshev inequality $(P(|Y - E\{Y\}| > a\sigma_Y) \leq \frac{1}{a^2})$,

$$P(|S_n - E\{S_n\}| > \epsilon) \leq \frac{\text{var}(S_n)}{\epsilon^2}.$$

But $E\{S_n\} = E\{X\}$ and $\text{var}(S_n) = \sigma_X^2/n$, so that

$$P(|S_n - E\{S_n\}| > \epsilon) \leq \frac{\sigma_X^2}{n\epsilon^2} \to 0 \text{ as } n \to \infty$$

Confidence Intervals

- Given ϵ, $\delta > 0$, how large should the number of samples n be such that

$$P(|S_n - E\{X\}| \leq \epsilon) \geq 1 - \delta\,,$$

i.e., with probability $\geq 1 - \delta$, S_n is within $\pm\epsilon$ of $E\{X\}$?

- Let's use the CLT to find an estimate of n. Consider

$$\begin{aligned} P(|S_n - E\{X\}| \leq \epsilon) &= P(\Big|\frac{1}{n}\sum_{i=1}^{n}(X_i - E\{X\})\Big| \leq \epsilon) \\ &= P(\Big|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\frac{X_i - E\{X\}}{\sigma_X})\Big| \leq \frac{\epsilon\sqrt{n}}{\sigma_X}) \approx P(|Y| \leq \frac{\epsilon\sqrt{n}}{\sigma_X}) \\ &= 1 - 2Q(\frac{\epsilon\sqrt{n}}{\sigma_X}) \end{aligned}$$

where $Z_n = Y \sim \mathcal{N}(0,1)$ and $Q(.)$ is Marcum's Q function defined as $Q(d) = \int_d^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$.

Examples

- Example 1. If $\epsilon = 0.1\sigma_X$ and $\delta = 0.001$, how many samples do we need to ensure that S_n is within $\pm 0.1\sigma_X$ of $E\{X\}$ with probability (confidence) of 0.999 or better, independent of the distribution of X?
  Solution using CLT: We set

$$2Q(0.1\sqrt{n}) = 0.001 \;\Rightarrow\; 0.1\sqrt{n} = 3.2905 \;\Rightarrow\; n = 1083$$

  [MATLAB communications toolbox has functions qfunc and qfuncinv. For $Q(d)$ use qfunc(d) and to find d such that $Q(d) = \delta$ use qfuncinv(δ).]

- The Pollster's Problem. Let p be the fraction of people that intend to vote for Candidate A against Candidate B. A pollster polls n people. For the ith person polled define the r.v.

$$X_i = \begin{cases} 1 & \text{prefers Candidate A} \\ 0 & \text{otherwise} \end{cases}$$

Let $S_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample mean of the X_i's, i.e., the fraction of people polled that prefer Candidate A. The pollster wants to choose n large enough to ensure that $P(|S_n - p| \leq 0.01) \geq 0.95$.
Solution using CLT: $P(|S_n - p| \leq 0.01) = P(|Z_n| \leq \frac{0.01\sqrt{n}}{\sigma_X})$. What is σ_X? Since $X \sim \mathit{Bernoulli}(p)$, $\sigma_X^2 = p(1-p) \leq 0.25$ for $p \in [0,1]$. Thus, $\sigma_X \leq 0.5$ and
$P(|S_n - p| \leq 0.01) \geq P(|Z_n| \leq \frac{0.01\sqrt{n}}{0.5}) = 1 - 2Q(0.02\sqrt{n}) = 0.95$
$\Rightarrow Q(0.02\sqrt{n}) = 0.025 \Rightarrow 0.02\sqrt{n} = 1.96$, which leads to $n = 9604$

Convergence of Random Sequences

- Non-random sequences Definition A sequence of real numbers $\{x_n\}$, $(n = 1, 2, \cdots)$, converges to a real number x if given any $\epsilon > 0$, there exists an integer N such that

$$|x_n - x| < \epsilon \text{ for every } n > N.$$

  The above convergence is denoted as $x_n \to x$ as $n \to \infty$, or $\lim_{n\to\infty} x_n = x$.
- Cauchy Criterion $\{x_n\}$ converges to a limit if and only if $|x_n - x_m| \to 0$ as $n, m \to \infty$. That is, given any $\epsilon > 0$, there exists an integer N such that

$$|x_n - x_m| < \epsilon \text{ for every } n, m > N.$$

- The Cauchy criterion is useful in checking convergence of a sequence without knowing the limit point.

- Example Suppose $x_n = \frac{1}{n}$, $n = 1, 2, \cdots$. It is easy to guess that its limit point is 0. Apply the definition: pick $\epsilon > 0$. Then $|x_n - 0| = |\frac{1}{n}| < \epsilon$ if $n > 1/\epsilon$. Pick $N = \lceil 1/\epsilon \rceil$ where $\lceil a \rceil$ denotes the smallest integer $\geq a$. Then for any $n > N = \lceil 1/\epsilon \rceil$,

$$|x_n - 0| = \frac{1}{n} < \frac{1}{N} \leq \epsilon\,.$$

For example, if $\epsilon = 0.01$, then $N \geq 100$ works. If $\epsilon = 0.003$, we take $N = \lceil 333.33 \rceil = 334$.

- Cauchy Criterion $|x_n - x_m| = |\frac{1}{n} - \frac{1}{m}| \leq \frac{1}{n} + \frac{1}{m} < \epsilon$ if $n, m > 2/\epsilon$. So pick $N = \lceil 2/\epsilon \rceil$. For example, if $\epsilon = 0.01$, then $N \geq 200$ works. If $\epsilon = 0.003$, we take $N = \lceil 666.66 \rceil = 667$.

Convergence of Random Sequences

- Definition: Mean-Square Convergence A sequence of real random variables $\{X_n\}$, $(n = 1, 2, \cdots)$, converges in the mean-square sense to a random variable X if $E\{|X_n - X|^2\} \to 0$ as $n \to \infty$. That is, given any $\epsilon > 0$, there exists an integer N such that

$$E\{|X_n - X|^2\} < \epsilon \text{ for every } n > N.$$

  The sequence $\{X_n\}$ and X are defined on the same probability space.
- The above convergence is denoted as $\lim_{n\to\infty} X_n \overset{m.s.}{=} X$, or $X_n \overset{m.s.}{\to} X$.
- The above convergence can be reformulated in terms of the Cauchy criterion when the limit is unknown. That is, given any $\epsilon > 0$, there exists an integer N such that

$$E\{|X_n - X_m|^2\} < \epsilon \text{ for every } n, m > N.$$

- Definition: Almost Sure Convergence A sequence of real random variables $\{X_n\}$, $(n = 1, 2, \cdots)$, converges almost surely to a random variable X if $|X_n(\omega) - X(\omega)| \to 0$ as $n \to \infty$ for every $\omega \in A$ such that $P(A) = 1$

  The above convergence is denoted as $X_n \overset{a.s.}{\to} X$ or $X_n \overset{w.p.1}{\to} X$ where w.p.1 stands for "with probability one."

- Example Pick $\Omega = [0, 1]$, and define $X_n = \exp\left(-n^2(\omega - \frac{1}{n})\right)$. For any subset A of Ω, we define $P(A) =$ length of A, equivalently, ω is uniformly distributed over $[0, 1]$. We have

$$\lim_{n\to\infty} X_n(\omega) = \lim_{n\to\infty} e^{-n^2(\omega - \frac{1}{n})} = \begin{cases} 0 & \text{for } 0 < \omega \le 1 \\ \infty & \text{for } \omega = 0 \end{cases}$$

Since $P(0 < \omega \le 1) = 1$, it follows that $X_n \overset{a.s.}{\to} 0$. But $X_n \overset{m.s.}{\not\to} 0$:

$$E\{|X_n - 0|^2\} = E\{e^{-2n^2(\omega - \frac{1}{n})}\} = e^{2n} \int_0^1 e^{-2n^2 y}\, dy$$

$$= e^{2n} \frac{e^{-2n^2 y}}{-2n^2}\Big|_{y=0}^{1} = \frac{e^{2n}}{2n^2}\left(-e^{-2n^2} + 1\right) \to \frac{e^{2n}}{2n^2} \to \infty$$

- Example: Sample Mean To estimate the mean, we collect $X_1, X_2, \cdots, X_n$ i.i.d. samples drawn according to the same distribution as X and compute the sample mean

$$S_n = \frac{1}{n} \sum_{i=1}^{n} X_i \, .$$

  It was shown earlier (slide 11) that $E\{|S_n - E\{X\}|^2\} = \sigma_X^2/n \to 0$. Hence $S_n \overset{m.s.}{\to} E\{X\}$.

- We also have $S_n \overset{a.s.}{\to} E\{X\}$ if $E\{X^4\} < \infty$, but to prove this fact we need lot more effort (see Example 14.15 in Gubner).

- Definition: Convergence in Distribution A sequence of real random variables $\{X_n\}$, $(n = 1, 2, \cdots)$, with CDF $F_n(x)$, converges in distribution to a random variable X with CDF $F(x)$ if $\lim_{n\to\infty} F_n(x) = F(x)$ for any x at which $F(x)$ is continuous.

  The above convergence is denoted as $\lim_{n\to\infty} X_n \overset{d}{=} X$ or $X_n \xrightarrow{d} X$.
- For convergence in distribution, the random variables $\{X_n\}$ and X are not necessarily defined on the same probability space, unlike other types of convergence for which same probability space is required.
- Example Suppose X_n is uniformly distributed over $[-\frac{1}{n}, \frac{1}{n}]$. We have

$$F_n(x) = \begin{cases} 0 & \text{for } x < -\frac{1}{n} \\ \frac{nx}{2} + 0.5 & \text{for } -\frac{1}{n} \leq x \leq \frac{1}{n} \\ 1 & \text{for } \frac{1}{n} < x \end{cases}$$

  Then

$$\lim_{n\to\infty} F_n(x) = \begin{cases} 0 & \text{for } x < 0 \\ 0.5 & \text{for } x = 0 \\ 1 & \text{for } x > 0 \end{cases}$$

- Example (continued) We have $F_n(0) = 0.5$ for any $n \geq 1$. Suppose random variable X has CDF $F(x) = u(x)$, i.e.,

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 & \text{for } x \geq 0 \end{cases}$$

$F(x)$ is continuous for every x except $x = 0$. Since $F_n(x) \to F(x)$ for every x except $x = 0$, it follows that $X_n \xrightarrow{d} X$.